

Time: 180 Minutes

Max. Marks: 50

- Instruction:**
- 1. In case of any doubt, write your assumptions, write it clearly and continue.**
 - 2. There are 06 questions printed on the paper. Attempt any 5 questions.**
 - 3. Please write all sub-parts of a question in a sequence.**

- Q1. (a) Which loss function is used for a classification problem? Mathematically define it. 02
 (b) Design a McCulloch Pitts neuron to model a 3-Bit logical OR problem. 02
 (c) Explain the weight updating equations for the following algorithms: 04
 i) Adam ii) NAG
 (d) What are the characteristics of a sequential learning problem? 02
- Q2. (a) Which of the following is/are true? 04
 a. Complex models has high bias and low variance.
 b. In overfitting train error is low while test error is high.
 c. Both Bias and Variance contribute to error.
 d. Pooling is used to detect edges.
 (b) Consider a convolutional neural network block whose input size is $128 \times 128 \times 16$. The block consists of the following layers: 06
 • A convolutional layer 32 filters with height and width 4 and 0 padding which has both a weight and a bias (i.e. CONV4-32)
 • A 4×4 max-pooling layer with stride 2 and 0 padding (i.e. POOL4)
 • A convolutional layer 16 filters with height and width 2 and 0 padding which has both a weight and a bias (i.e. CONV2-16)
 Compute the output activation volume dimensions and number of parameters of the layers. You can write the activation shapes in the format ($H; W; C$ where $H; W; C$ are the *height, width, and channel* dimensions, respectively.
 i) What is the output activation volume dimensions and number of parameters for CONV4-32?
 ii) What is the output activation volume dimensions and number of parameters for POOL4?
 iii) What is the output activation volume dimensions and number of parameters for CONV2-16?
- Q3. (a) Write a short note on LSTM by clearly mentioning the equations for each of the gates. 05
 (b) If the sets P and N are finite and linearly separable, prove that the perceptron learning algorithm updates the weight vector w_t a finite number of times such that the two sets are separated. 05
- Q4. (a) Explain the ReLU activation function by computing its derivative? What is the problem in ReLU activation function? Explain how Leaky ReLU avoids the problem of ReLU activation function? 04
 (b) i) A 1-hidden layer neural network with 5 neurons, 6 inputs and 2 outputs with bias at neuron will have how many total parameters? [02 Marks] 06
 ii) How does splitting a dataset into train, validation and test sets help identify overfitting? [02 Marks]
 iii) You come across a nonlinear function that passes 1 if its input is nonnegative, else evaluates to 0, i.e.

$$f(x) = \begin{cases} 1 & | x \geq 0 \\ 0 & | x < 0 \end{cases}$$

 A friend recommends you use this non-linearity in your convolutional neural network with the SGD optimizer. Would you follow their advice? Why or why not? [02 Marks]
- Q5. (a) Draw the architecture of LeNet by mentioning the hyper-parameters of each layer. 03
 (b) Design a multilayer perceptron model to map the XOR function with 3 variables with only a single hidden layer and an output layer. 02
 (c) Prove that moving in the opposite direction to the function Gradient minimizes the function. Explain the Backpropagation algorithm using gradient descent update rule. 05

- Q6. (a) The convolution operation is applied on an input image (as in Fig.1) using a filter as in Fig.2 with stride 2. What will be the output image? If padding is to be used on the convoluted image to make its dimension same as input image what will be the padding size? **05**

0	2	0	4	0	0
0	0	5	0	1	0
0	1	0	3	0	1
0	2	0	0	4	0
1	0	0	5	0	1
0	2	1	0	3	0

Fig. 1 Input Image

1	0	-1
2	0	-2
1	0	-1

Fig.2 Filter

- (b) What do you mean by transfer learning? Explain different cases of transfer learning. **03**
(c) What is the advantages of using skip connections in deep learning models? **02**

*****End of Question Paper*****